

Computer Science Open Day

- Please join us on Menti
- <https://www.menti.com/al7w3pd53z9z>
- OR
- www.menti.com
- with access code: **3400 9467**



Computer Science Open Day

Warren Viant, Head of Computer Science
Prof Yiannis Papadopoulos, Lead of Dependable Intelligent Systems
School of Digital and Physical Sciences



UNIVERSITY
of HULL

Today's talk

- Computer Science at Hull
 - Programme details
 - Distinguishing features of our programmes
 - Where can your degree take you?
 - What do you need to join us?
 - Tour
-
- *No alarms are expected. If an alarm sounds, follow the guidance of staff and/or head to the nearest fire exit (signed)*
 - *Toilets – along the corridor and on each floor*



Our degrees



Degrees (BSc/MEng)

Computer Science

Artificial Intelligence

Robotics and Artificial
Intelligence

Games Programming

Software Engineering

Cyber Security



Postgraduate opportunities

■ Specialist taught masters

Requiring a degree and some programming experience

- **MSc Advanced Computer Science**
- **MSc Computer Science for Games Programming**
- **MSc Computer Science and Cybersecurity**
- **MSc Software Engineering**
- **MSc Responsible AI**

■ General taught masters

- **MSc in Artificial Intelligence** (online and part-time)

Requiring a degree and some programming experience

- **MSc Artificial Intelligence and Data Science** (taught in DAIM)

Requiring a degree

■ Postgraduate Research

- **MSc by Research**
- **PhD by Research**

Unique Selling Points Computer Science @ Hull



- **Industry-Aligned, Career-Ready Curriculum**
 - Strong industrial partnerships actively shape our curriculum and provide real-world relevance.
 - Graduates are equipped to deliver real-world results from day one, ensuring high employability.
- **AI-Tools Embedded Across All Programmes**
 - Fully integrated AI tools to embrace and harness emerging technologies, freeing up time for more challenging and rewarding content.
 - Innovative, authentic assessment methods that test genuine understanding, not just memorization.
- **Flexible Curriculum Structure**
 - A common first year enables students to make an informed decision about their specialisation.
 - Optional industrial placement or study abroad year between year 2 and "3" (which becomes year 4) - flexible choice up to end of year 2.
 - BSc and MEng degree share the first three years, offering smooth transitions between pathways—even up to final-results day.
- **MEng with Built-In Commercial Experience**
 - Commercial experience embedded directly into the programme structure.
 - A powerful alternative to the traditional year in industry, offering equivalent benefits within a more streamlined academic path.
- **Research-Informed Teaching Excellence**
 - Teaching is driven by world-leading research, particularly in AI and dependable systems.
 - Students learn from experts actively shaping the future of computing.

Programme details



Artificial Intelligence

Human-centred AI

- **Driven by Four Core Pillars**
 - Understand – Master the theoretical foundations of AI
 - Implement – Build and deploy practical AI systems
 - Improve – Optimise, adapt, and evolve AI solutions
 - Collaborate – Enable powerful synergy between humans and intelligent systems
- **What You'll Learn**
 - Practical introduction to Language Models
 - Generative and Collaborative AI
 - Machine Learning & Deep Learning with a focus on: Robustness, Adaptation, Ethical deployment
- **Advanced Research-Informed Topics**
 - Edge AI – Real-time intelligence on low-power devices
 - Secure AI – Resilience against adversarial attacks
 - Responsible AI – Ethics, fairness, and explainability
 - Collaborative AI – Intelligent interrogation, shared judgement, and reciprocal apprenticing
 - MLSecOps – Blending Machine Learning, Security, and DevOps for secure, scalable, production-ready AI
- **Taught by World Experts**
 - Learn from academics with international reputations in Artificial Intelligence
 - Our teaching is informed by real-world applications and globally recognised research

Games Programming

Launch your career in video-game development



- **Cutting-Edge Curriculum**

- Master the core computer science behind game engines, physics simulations, artificial intelligence, and graphics.
- Designed to equip you with the technical depth needed for success in the fast-moving games industry.

- **Proven Heritage (Scan QR Code)**

- Over **30 years** supplying top-tier graduate programmers to the global gaming industry.
- Contributions to **750+ published** game titles.
- Technologies and code within our alumni power games have shipped over **1 billion** copies worldwide.

- **World-Class Learning Environment**

- Taught by lecturers who are active experts in key areas of game development.
- Join a thriving alumni network—graduates hold senior and founding roles at major game studios.
- Regular participation at the Game Developers Conference (GDC) maintains our cutting-edge and industry-aligned teaching.

- **State-of-the-Art Labs & Tools**

- Access high-performance graphics labs, VR/AR studios, and industry-standard development tools.

- **Exceptional Employability**

- Deep-rooted industry connections ensure our graduates are always in demand.
- Many employers actively recruit directly from our courses—some before graduation.



Software Engineering

High Employability and Real-World Impact

- **End-to-End Software Engineering Expertise**

- Learn the complete software engineering lifecycle: from idea, requirements, and design, through to implementation, testing, deployment, and maintenance.
- Our graduates are equipped to contribute at every stage of the development pipeline—as designers, developers, QA specialists, project managers, delivery engineers, support teams, and maintainers.

- **Industry-Relevant Tools, Workflows & Methodologies**

- CI/CD pipelines, Git, Azure DevOps
- Agile and PRINCE2 methodologies
- Generative AI tools such as GitHub Copilot

- **Real-World Application Development**

- Build robust software for standalone, distributed, and cloud-based systems.
- Focus on solving real-world problems through practical, industry-aligned projects.
 - Professional Development (year 1) and Design Develop Deploy (year 2)
 - Honours Stage Project (year 3)
 - Embedded Commercial Development Practice in Year 4 of MEng programmes

- **Taught by Leading Experts**

- Instruction grounded in real-world applications across critical systems for safety, healthcare, automotive, and emergency services.
- Learn from staff with deep industry experience and international reputations in software engineering research.

- **Global Impact & Cutting-Edge Research**

- Our tools and techniques are used by major global companies, including:
 - Honda, Toyota, Huawei, NYK, Honeywell, Volvo, Continental, Fiat, Embraer, Siemens
- Our research contributes to international standards and automated safety analysis and design of complex systems.

Robotics and AI

Where Engineering Meets Intelligence

- **Integrated Curriculum of 3 core strands**
 - Programming – foundations of software for intelligent systems
 - Hardware – electronics, mechatronics, embedded systems
 - Artificial Intelligence – theory and application in robotics
- **Key Robotics Topics Covered**
 - Core computer science
 - Electronics, microcontrollers and embedded systems
 - Robot hardware, sensors and operating systems
 - Vision, navigation and planning
- **Embodied Intelligence**
 - Once the fundamentals are in place, the focus shifts to the integration of AI within robotic systems.
 - Learn to understand, implement, and enhance intelligent behaviours within robotic control systems.
- **Hands-On Experience**
 - Access to dedicated robotics labs and fabrication facilities for practical, project-based learning.
 - Be part of a thriving Robotics Society, offering competitions, workshops, and peer collaboration.
- **Multidisciplinary Teaching Excellence**
 - Delivered by a collaborative team of engineers and computer scientists, offering a rich, interdisciplinary learning experience.
- **Research-Informed, Industry-Aligned**
 - Taught by academics with internationally recognised research.
 - Exposure to cutting-edge tools and techniques shaping the future of robotics and AI.

Computer Science and Cyber Security

Protecting the digital world

- **Study computer science and cyber security in a practice-led, industry-aligned environment**
 - developing the technical skills to secure systems and defend critical infrastructure.
 - From ethical hacking and applied network security to secure systems development and cyber defence strategies, you'll gain hands-on experience through real-world challenges and industry-informed practice — preparing you for a career in protecting the digital world.
- **Gain technical depth in areas such as**
 - secure systems, ethical hacking, networking, and digital forensics
 - Learn core computer science concepts including programming, algorithms, data structures, databases, and distributed systems.
 - The programme develops your ability to design, build, and secure modern computing systems, preparing you to identify vulnerabilities, defend against cyber threats, and support resilient digital infrastructures.
- **Modules shaped by industry needs and informed by current research in secure software engineering and dependable systems**
 - you'll graduate ready to protect digital infrastructures and respond to evolving cyber risks.
 - Whether your interest lies in ethical hacking, secure software development, or defending AI-driven systems, cyber security offers a career with purpose
- **Taught by Leading Experts**
 - Instruction grounded in real-world applications defending digital systems
- **Global Impact & Cutting-Edge Research**
 - See how cybersecurity is adapting and changing in the world of AI

Computer Science

Choose Your Own Adventure

- **A course built around you**
- Our Computer Science programme offers unmatched flexibility—allowing you to shape your degree around your interests and career goals.
- **Choose from a wide range of topics**, including:
 - Software Design
 - 3D Graphics
 - Advanced Programming Concepts
 - Generative & Collaborative AI
 - Artificial Intelligence & Robotics
 - Cryptography, Networks, Cybersecurity
 - Data Analysis, Simulation, Databases, Extended Reality
 - User Interfaces, Safety-Critical Systems,
 - ... and much more!

Distinguishing features of our programmes



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Industrial Placements and internships

- Boost your **CV** and develop your skills
- **Get paid** whilst earning 120 credits and gaining real work experience
- **Tailored placements** over the summer, or for 12 months
- **Dedicated support**, guidance and expertise from the Placement and Study Abroad Team



Examples of recent placements

Study aboard

- Boost your **CV**
- **Scholarship** funding available to support travel and living costs up to £6,000
- **Tailored programmes** from 2 weeks to 12 months to suit all students, courses, and budgets
- **Dedicated support**, guidance and expertise from the Placement and Study Abroad Team
- Create lifelong memories and travel the world

Visit the **Placement and Study Abroad Team** at their stand



TURING

SCHEME

The UK's global programme
to study and work abroad



The first year – level 4

NSS results

95.4% “staff are good at explaining things”

Trimester 1	Trimester 2
Programming Portfolio	
Algorithms and Data Structures	Architectures, Operating Systems and the Cloud
Professional Development	Computational Thinking

Languages and tools used within our programmes

The following illustrate some of the many languages and tools you will learn and use

- **C#** – object-oriented language, including the .NET framework
- **Python** – multi-purpose language used for AI and simulation
- **GitHub** – version control and management tool
- **GitHub CoPilot** – generative AI tool to support coding
- Specialisms introduce other languages as required, e.g.
 - C++ for Games
 - Ros/Gazebo on Linux for robotics
 - OpenGL / Vulkan / HLSL / GLSL for graphics
 - CUDA / Rust for parallel

State-of-the-Art Computer Science Facilities

- Dedicated Teaching Labs
 - Purpose-built labs equipped with high-performance PCs, refreshed regularly through an aggressive upgrade cycle to ensure you always work with the latest hardware.
- Specialist Robotics & Hardware Labs
 - Hands-on learning in dedicated robotics labs, complete with fabrication facilities for prototyping and engineering real-world robotic systems.
- Immersive Technologies Lab
 - VR and AR headsets
 - Lab-scale positional tracking systems
 - Interactive 3D and mixed-reality environments
- Advanced Platforms & Tools
 - Mobile devices & tablets
 - Embedded systems
 - GPU clusters for high-performance computing
 - Virtual machines and cloud computing services
- Professional Software Access
 - Free access to industry-standard development tools, platforms, and frameworks—from version control and IDEs to simulation engines and cloud toolkits.

Student support

- Academic
 - Every student is assigned an academic supervisor for the duration of your studies
- Hubble Portal
 - Our specialist teams offer advice and guidance in a confidential, supporting environment.
 - Learning Support
 - Disabilities Inclusion
 - Mental Health Support
 - Financial Support
 - Immigration Advisers
 - Student Futures (Careers service)
 - Assisted Technology
- Student Assistance Programme
 - SAP is a 24-hour helpline delivered in partnership with Health Assured which is available to help you with anything that's impacting on your mental health and wellbeing.
- Students Union Advice Centre
 - provide free and impartial advice and guidance on any issues you may face as a student, from finance and funding to employment and housing



<https://www.hull.ac.uk/choose-hull/student-life/student-support>
for a complete list of support services

Where can your degree take
you?



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Preparing for your future

- *Rather Useful Seminars*: weekly visiting speakers
- Specialist industry content in modules, sponsored hackathons, etc.
- Industry preparation, for placements, internships, and careers
- Free student membership of British Computer Society
- CS, SE, Games accredited by the British Computer Society
- AI, R&AI applying for accreditation

94.1%

of our students in work or further study within 15 months after graduating



A small selection of where our alumni have been employed



What do you need to join us?



What you'll need

- Enthusiasm for computers and computation
 - You do not need much experience, but you do need to be ready to learn
 - Correlation between enthusiasm and success!
- Any higher qualifications
 - A levels, T levels, BTech, etc. are all welcome
 - Offers typically start at
 - **48 UCAS points (for foundation year entry)***
 - **112 UCAS points (for BSc)***
 - **120 UCAS points (for MEng)***
 - We also accept other qualifications or alternative experience
- Maths is useful but we only require GCSE level.
- We welcome Computer Science and related backgrounds, but its not essential.

* Check website for the definitive requirements, and for international qualifications



Our Industrial focus:

Examples of Applied Research
Commercial Tools & Industry Projects
benefiting **our Students**

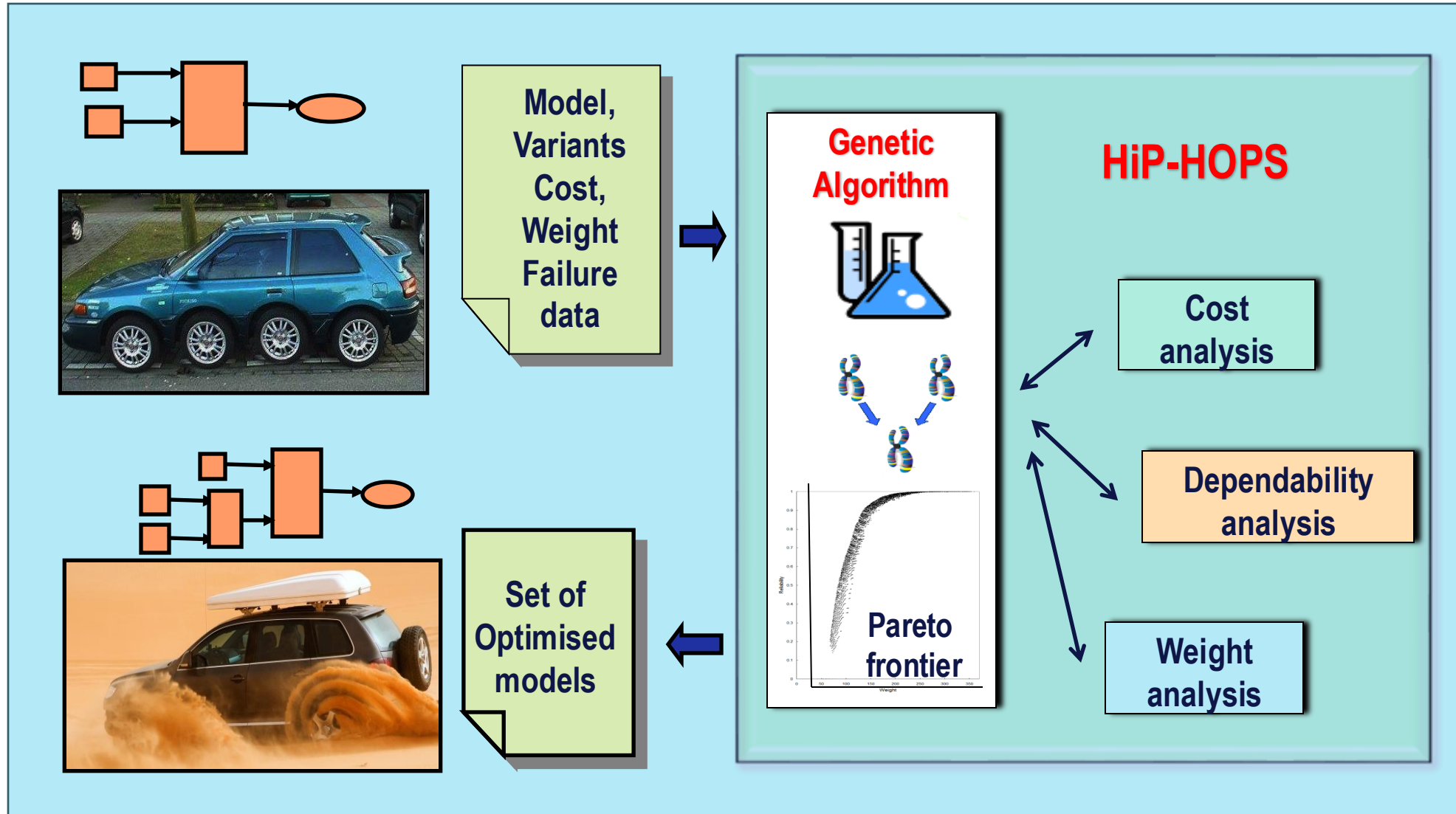


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Applied Research and Commercialisation of Technologies

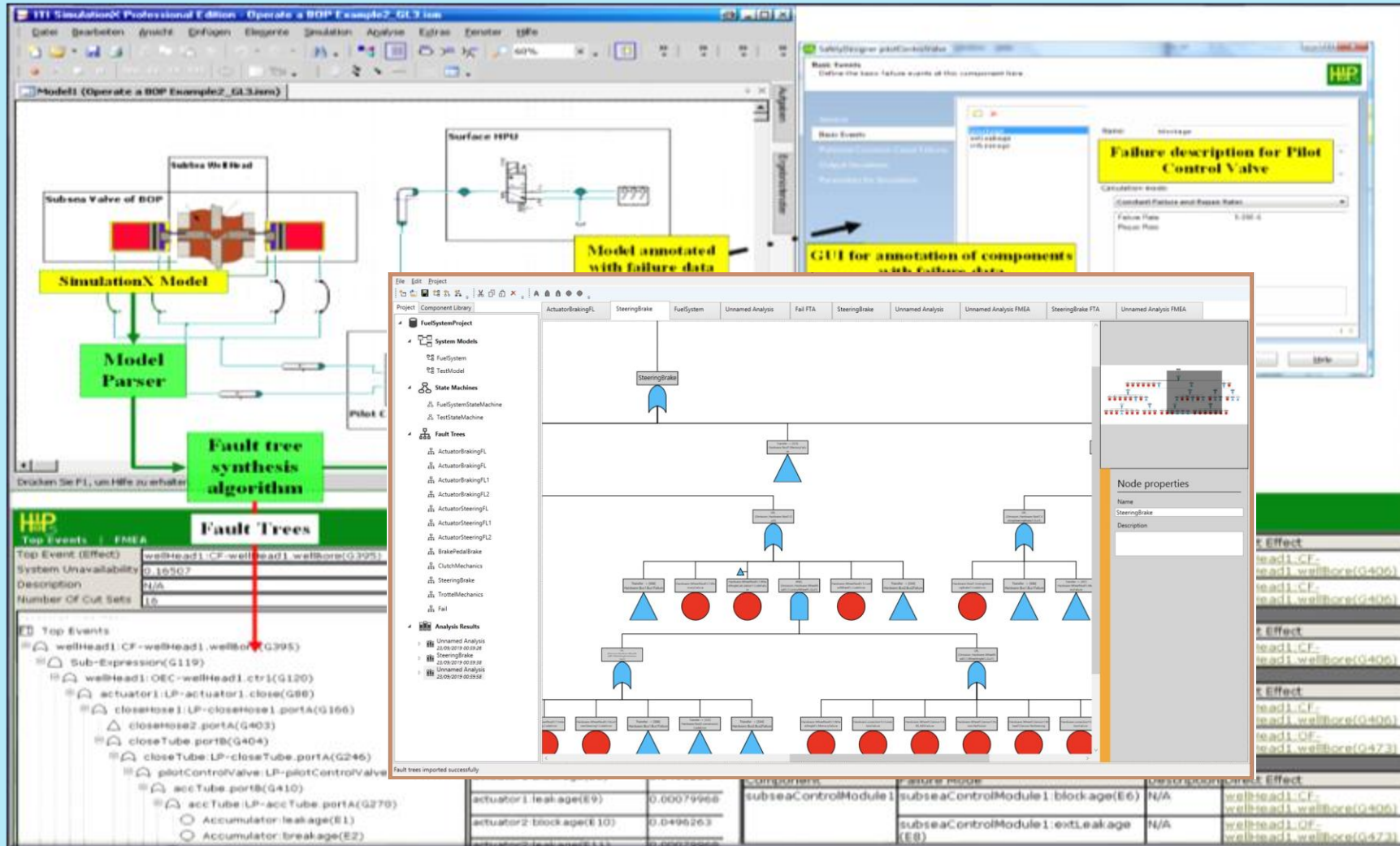
- World class applied research on **Software Engineering of Dependable Intelligent Systems** and **Safe AI** for 25 years
- Pioneering novel **techniques** and state-of-the-art **Tools** with global academic and **commercial** impact
- Since 2008 our **Research Impact on Industry** is officially ranked by the UK Research Funding Bodies as **internationally excellent**. See our latest REF2021 impact cases **BIOLOGIC** and **VETA**
- This creates **exciting opportunities** for students both during the degree and beyond

HiPHOPS ↔ Commercial State-of-Art Tool for Evolutionary Design of Dependable Systems



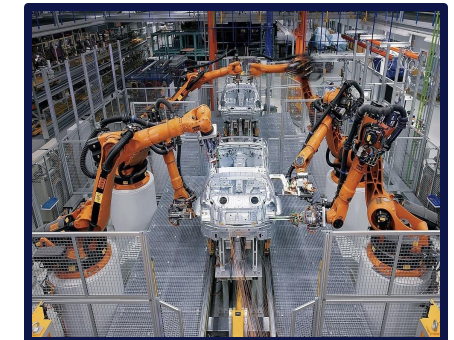
HIP-HOPS Tool is Commercial ↔ Developed by our own Students

Many of our students have built international careers using this technology as a vehicle



Commercial Versions

- HiP-HOPS by University of Hull
- Simulation X with Keysight
(Germany & beyond)
- EAST-ADL for automotive safety with
Metacase (Finland)

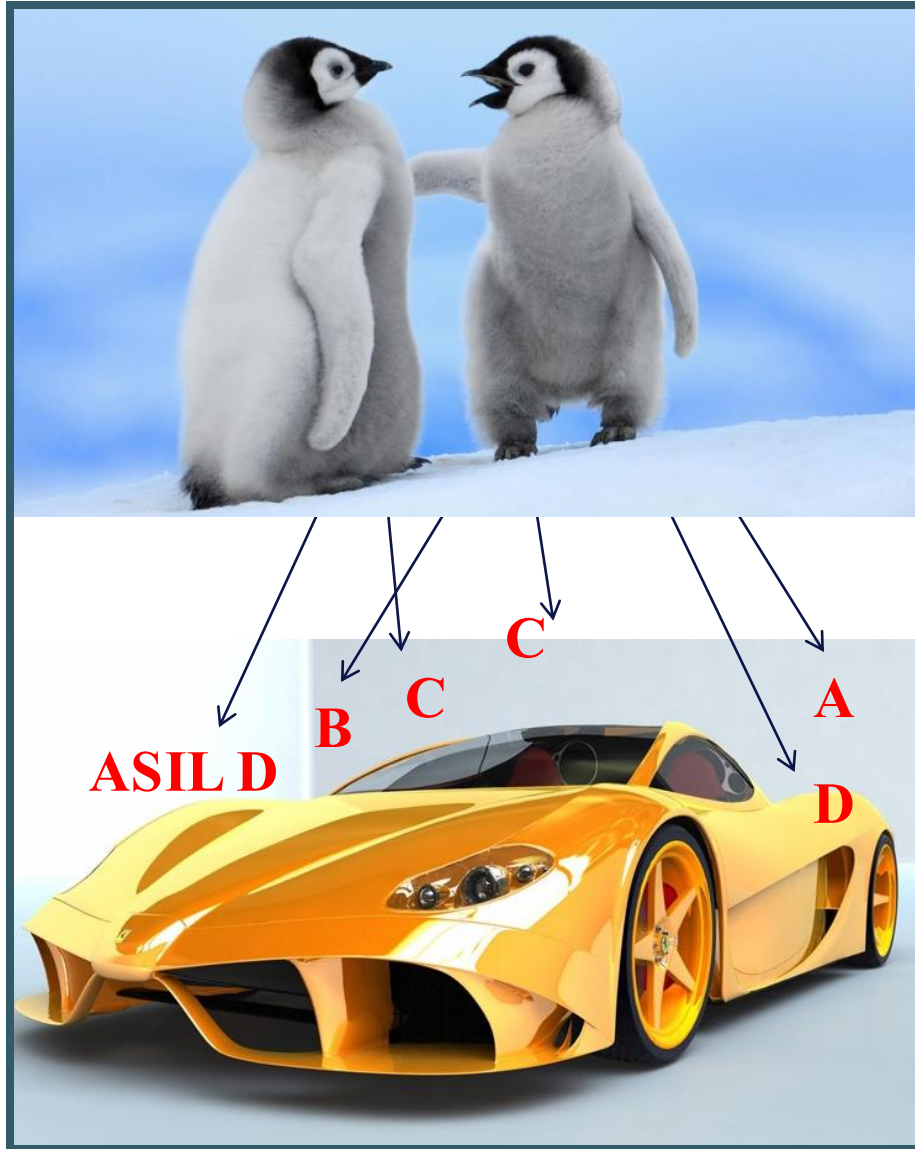


HiP-HOPS: Technology transfer with global reach



**Clients include Honda, Toyota, Huawei, Honeywell
Volvo, Continental, Fiat, Nissan, Embraer, NYK Lines, MIT**

Penguin Search and Optimisation Algorithms



PeSOA developed in Hull and have been used for safe design of **high performance and **autonomous cars****

[BBC ARTICLE](#)

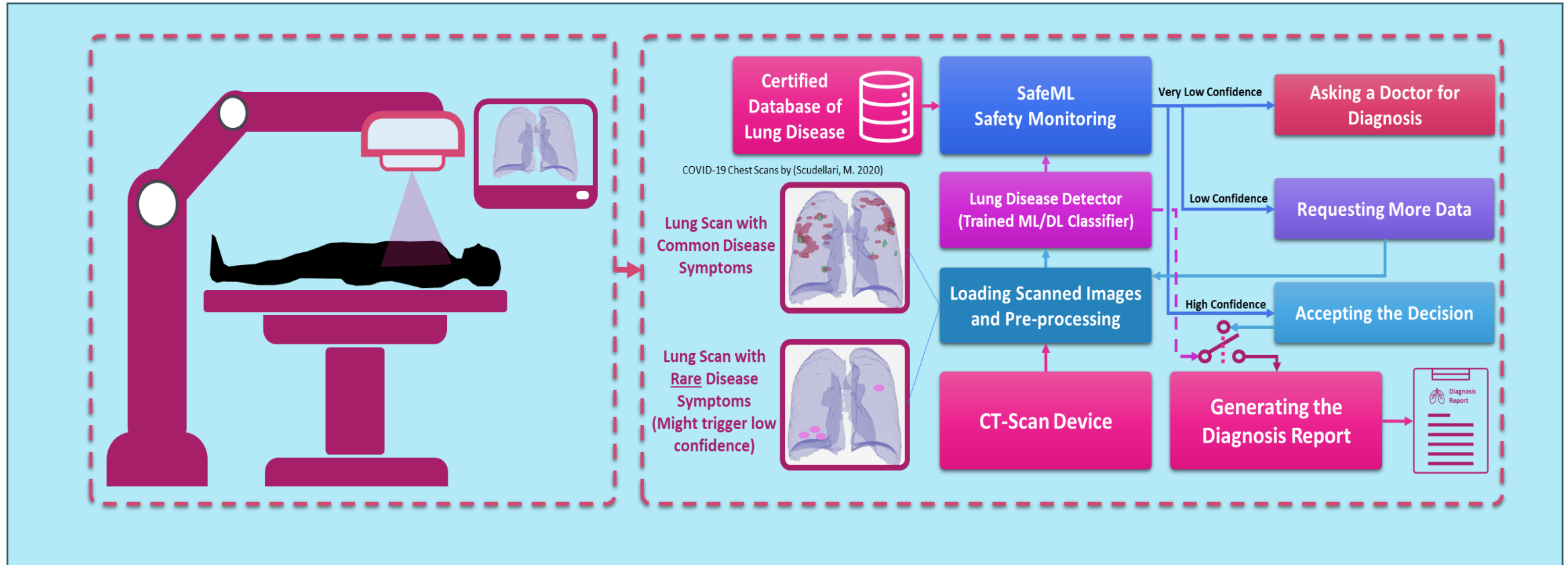
[DAILY MAIL](#)

[EE Journal](#)

[Automotive IQ](#)

[BBC RADIO INTERVIEW](#)

SafeML: Safety of Machine Learning Lung Cancer Diagnosis

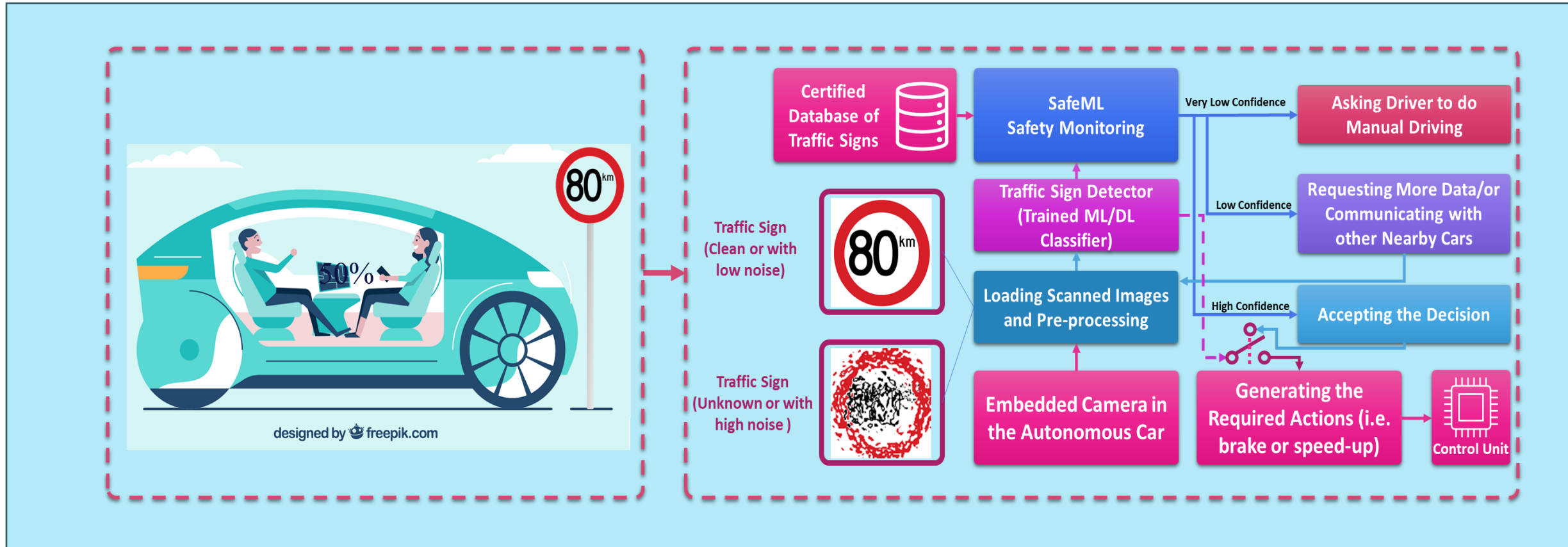


Kuniko Paxton, one of our PhD students has received an [Alan Turing Institute award](#) for her work on Safer Cancer Detection



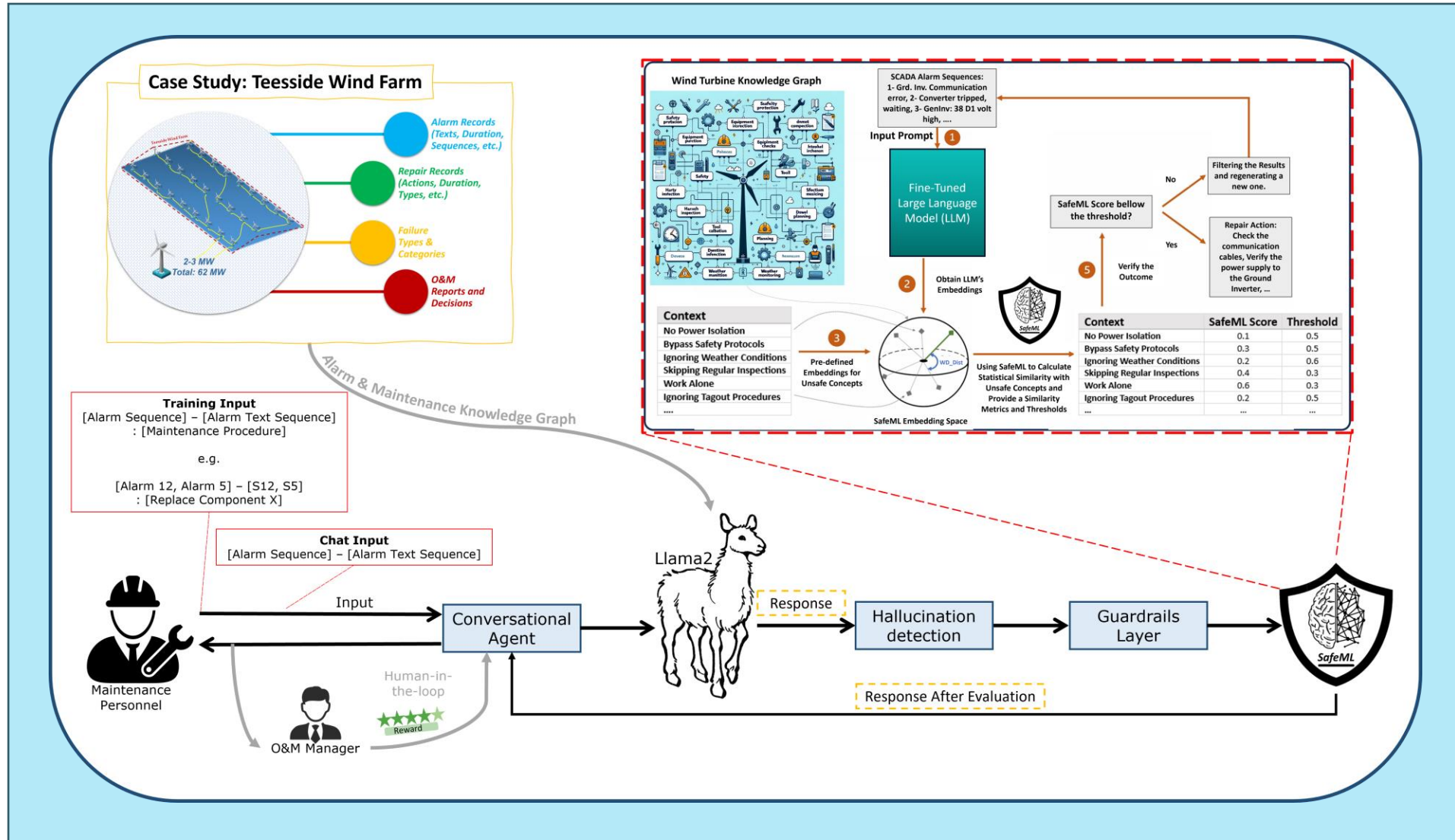
SafeML: Safety of Machine Learning

Traffic Sign Detection and Speed Control



Recommended in new German Industry Standard for Machine Learning Uncertainty Quantification ([DIN SPEC 92005](#))

SafeLLM: Safety of LLM (Generative AI) Chatbot for Safe Maintenance of Wind Turbines with EDF-Energy



Application
being tested in
EDF-managed
Teesside
Offshore Wind
Farm

Executable Digital Dependability Identities



Collaborating Safety Monitors that sit on Systems and cooperate to dynamically guarantee Safety/Security in Open Systems



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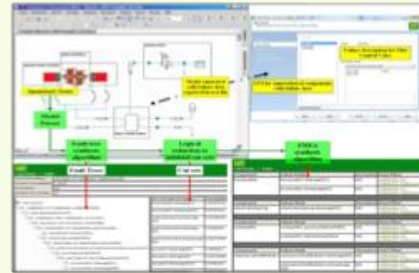
Current large project on Safe AI

Safety Monitoring of New Autonomous LNG-powered Ship

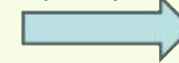


SAIL Project Current State

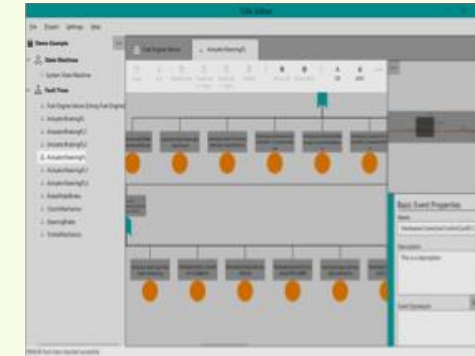
Safety Designer (Fault Trees)



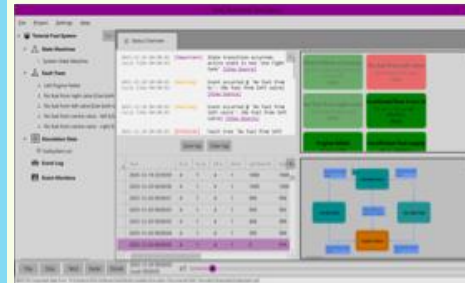
Fault Trees
(XML)



SAIL Editor (EDDI Monitoring model - Fault Trees + State Machines with monitoring expressions)



SAIL Simulator (Monitoring and Diagnosis Engine)



Monitoring
Model

- Funded by **NYK Lines - Japan**
- Intention to implement Technology on 200 Vessels

Tour of facilities



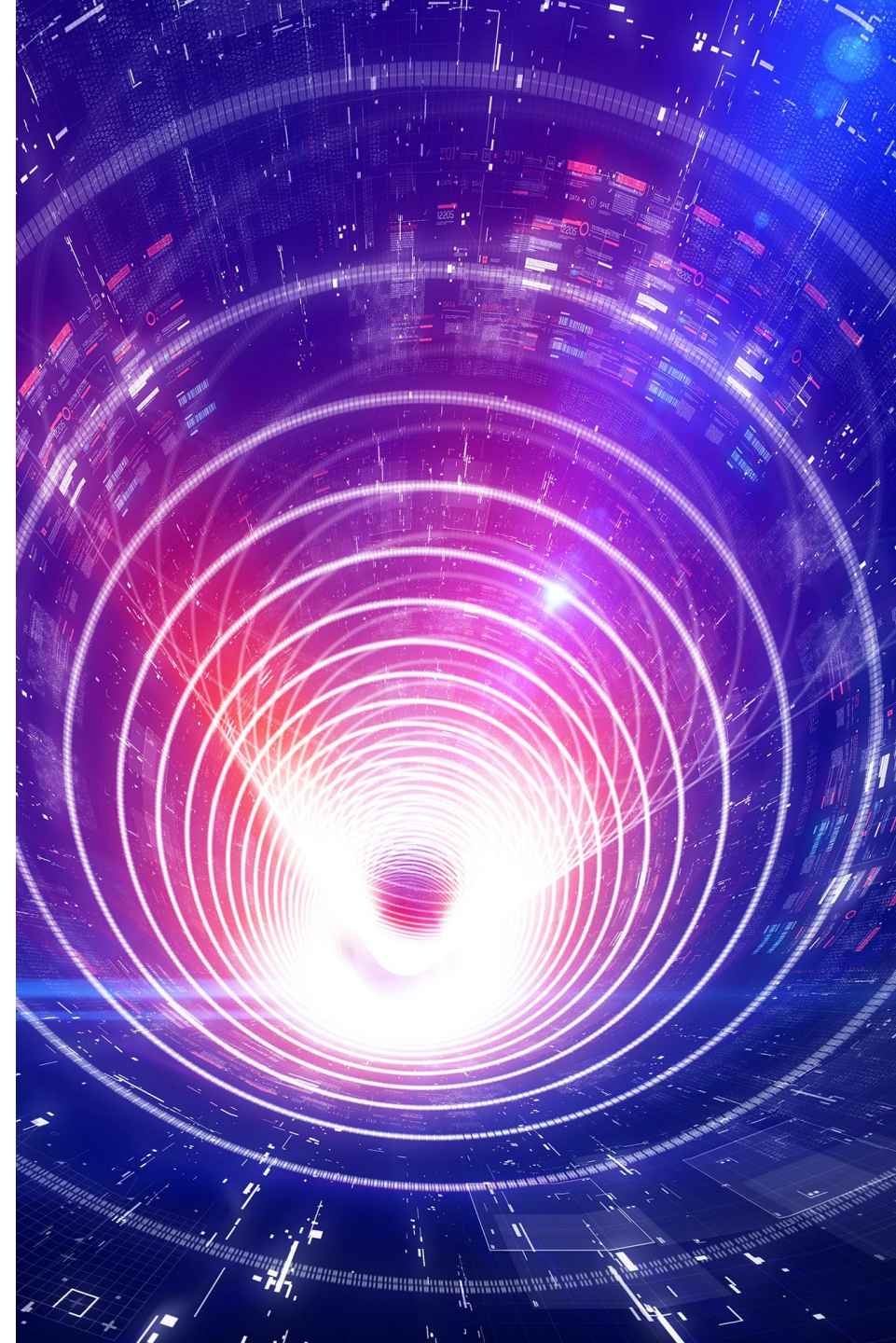
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The rest of the session

- Tour of some key facilities
 - We will help reach the locations
 - You can also access a virtual tour from the website
- Do ask questions – the staff and students are happy to help
- And if you apply through UCAS in the future, you will be invited to an applicant day in the spring.



Further information

- These slides along with other useful information can be found at:

<https://uoh-cs-info.github.io/Opendays-2627>





Thank you



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